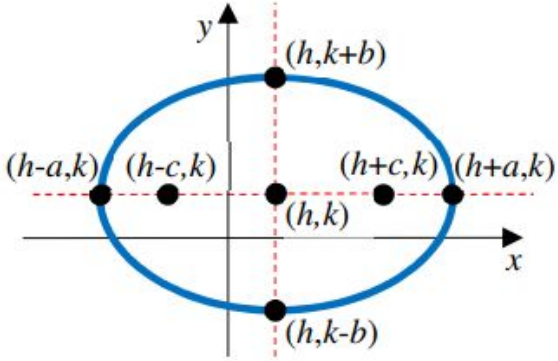
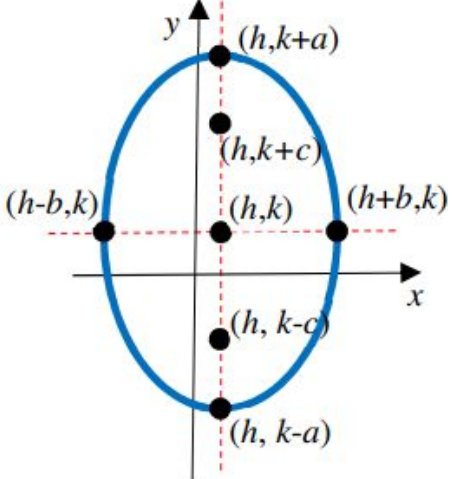
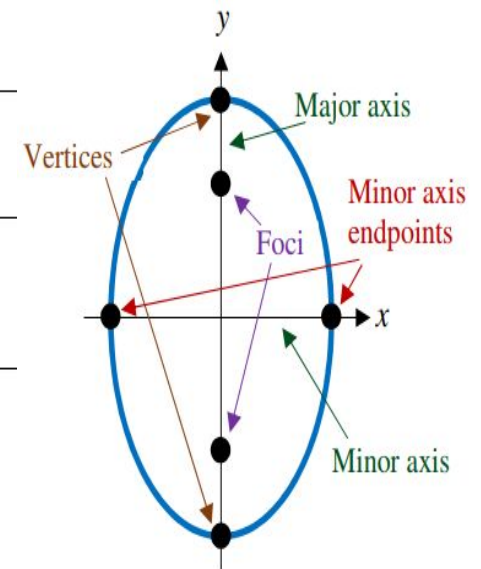


Conics

Prepared by
Shahed Alam

Ellipse

Major Axis	Horizontal	Vertical
Standard Equation	$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$	$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$
Vertices	$(h \pm a, k)$	$(h, k \pm a)$
Minor Axis Endpoints	$(h, k \pm b)$	$(h \pm b, k)$
Foci	$(h \pm c, k)$ where $b^2 = a^2 - c^2$	$(h, k \pm c)$ where $b^2 = a^2 - c^2$
Graph		



subject :

Date :

Ellipse foci $(\pm 4, 0)$, passing through $(-4, 1.8)$

horizontal ellipse since x values are changes

Centre $(h, k) = \emptyset$

Foci $(h \pm c, k) = (\pm 4, 0)$

$$\begin{array}{l|l} h+c=4 & h+c=4 \\ h-c=-4 & 2, 0+c=4 \\ \hline 2h=0 & c=4 \\ h=0 & k=0 \end{array}$$

$$\begin{aligned} c^2 &= a^2 - b^2 \\ a^2 - b^2 &= 4^2 \quad \text{--- eq (1)} \end{aligned}$$

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

$$\Rightarrow \frac{(-4-0)^2}{a^2} + \frac{(1.8-0)^2}{b^2} = 1$$

$$\Rightarrow \frac{16}{a^2} + \frac{1.8^2}{b^2} = 1 \quad \text{--- eq (2)}$$

LD $a^2 = 4^2 + b^2$

$$\frac{16}{4+b^2} + \frac{1.8^2}{b^2} = 1$$

$$\frac{16}{4+b^2} + \frac{1.8^2}{b^2} = 1$$

$$\Rightarrow 16b^2 + (1.8)^2 (4-b^2) = b^2 (4+b^2)$$

$$\Rightarrow 16b^2 + 12.96 - 3.24b^2 = 4b^2 + b^4$$

$$\Rightarrow b^4 - 8.76b^2 - 12.96 = 0$$

$$b^2 = 10.05, \quad b^2 = 1.286$$

We know $4 + b^2 = a^2$

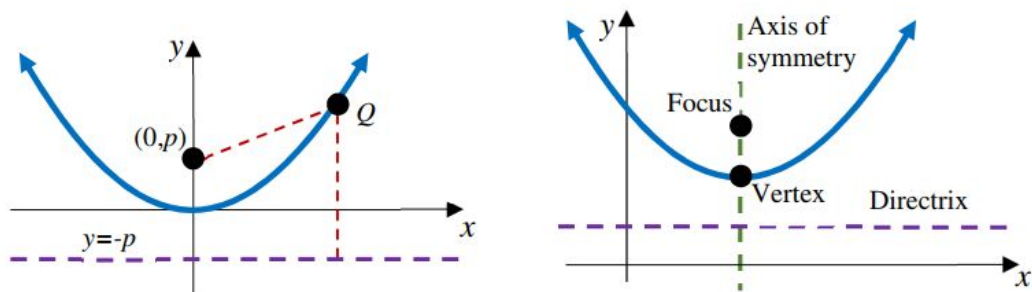
$$2) \quad 4 + b^2 = 10.05$$

$$2) \quad b^2 =$$

$$\Rightarrow 4 + 10.05 = a^2$$

$$\Rightarrow a^2 = 14.05$$

$$\frac{x^2}{14.05} + \frac{y^2}{10.05} = 1$$



	Horizontal	Vertical
Standard Equation	$(y - k)^2 = 4p(x - h)$	$(x - h)^2 = 4p(y - k)$
Vertex	(h, k)	(h, k)
Axis of symmetry	$y = k$	$x = h$
Directrix	$x = h - p$	$y = k - p$
Focus	$(h + p, k)$	$(h, k + p)$
Graph	An example with $p < 0$	An example with $p > 0$

Subject : _____



Date : _____

$$(y+3)^2 = 4(x-2)$$

Vertex $(2, -3)$

$$(y-k)^2 = 4p(x-h)$$

$$p = 1$$

Axis of symmetry $\Rightarrow y = k$
 $y = -3$

Since $p > 0$

Directrix $x = h - p$

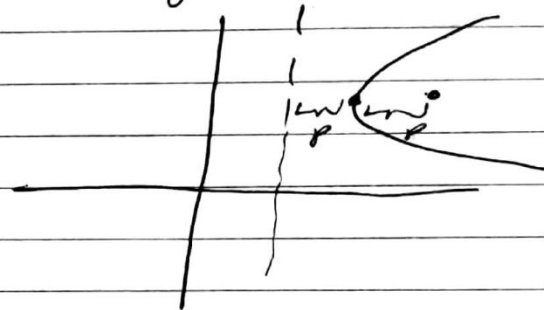
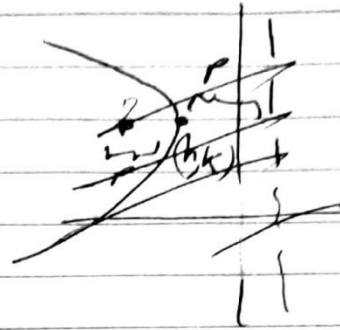
$$x = 2 - 1$$

$$\therefore x = 1$$

Focus $(h+p, k)$

$$(2+1, -3)$$

$$(3, -3)$$



Subject : _____

Date :

(20) Vertex at $(1, 3)$, focus at $(0, 3)$

Since the y coordinate values are ~~not~~ not changing it is a horizontal parabola.

$$\text{focus} \Rightarrow (h+p, k)$$

$$(0, 3)$$

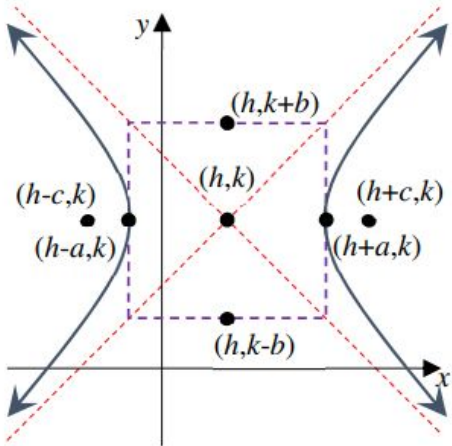
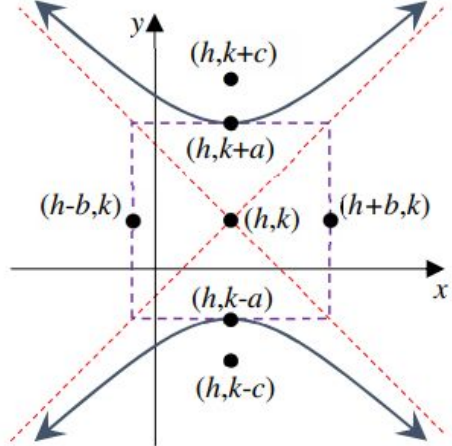
$$h+p=0$$

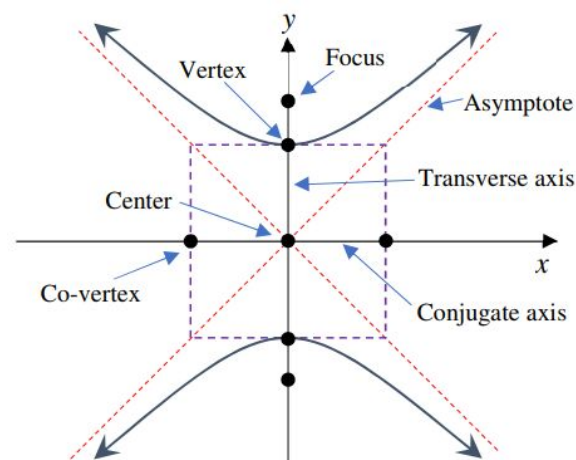
$$2) 1+p=0, \quad p=-1.$$

$$(y-3)^2 = -4(x-1)$$

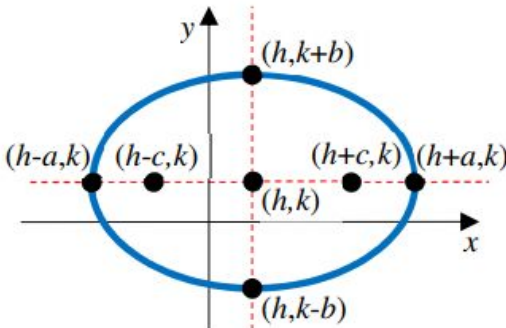
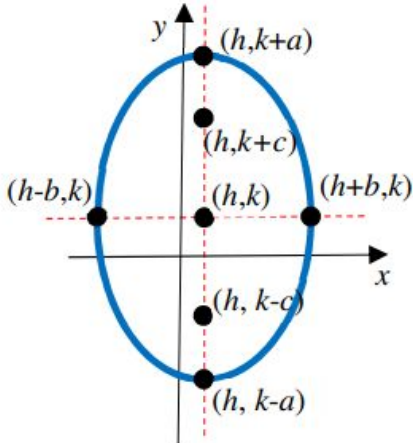
Example 5

Put the equation of the hyperbola $9x^2 + 18x - 4y^2 + 16y = 43$ in standard form. Find the center, vertices, length of the transverse axis, and the equations of the asymptotes. Sketch the graph, then check on a graphing utility.

Opens	Horizontally	Vertically
Standard Equation	$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$	$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$
Vertices	$(h \pm a, k)$	$(h, k \pm a)$
Foci	$(h \pm c, k)$ where $b^2 = c^2 - a^2$	$(h, k \pm c)$ where $b^2 = c^2 - a^2$
Asymptotes	$y - k = \pm \frac{b}{a}(x - h)$	$y - k = \pm \frac{a}{b}(x - h)$
Construction Rectangle Vertices	$(h \pm a, k \pm b)$	$(h \pm b, k \pm a)$
Graph		



Find the standard form of the equation for an ellipse centered at $(-2,1)$, a vertex at $(-2,4)$ and passing through the point $(0,1)$.

Major Axis	Horizontal	Vertical
Standard Equation	$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$	$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$
Vertices	$(h \pm a, k)$	$(h, k \pm a)$
Minor Axis Endpoints	$(h, k \pm b)$	$(h \pm b, k)$
Foci	$(h \pm c, k)$ where $b^2 = a^2 - c^2$	$(h, k \pm c)$ where $b^2 = a^2 - c^2$
Graph		

Major Axis	Horizontal	Vertical
Standard Equation	$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$	$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$
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Minor Axis Endpoints	$(h, k \pm b)$	$(h \pm b, k)$
Foci	$(h \pm c, k)$ where $b^2 = a^2 - c^2$	$(h, k \pm c)$ where $b^2 = a^2 - c^2$
Graph	